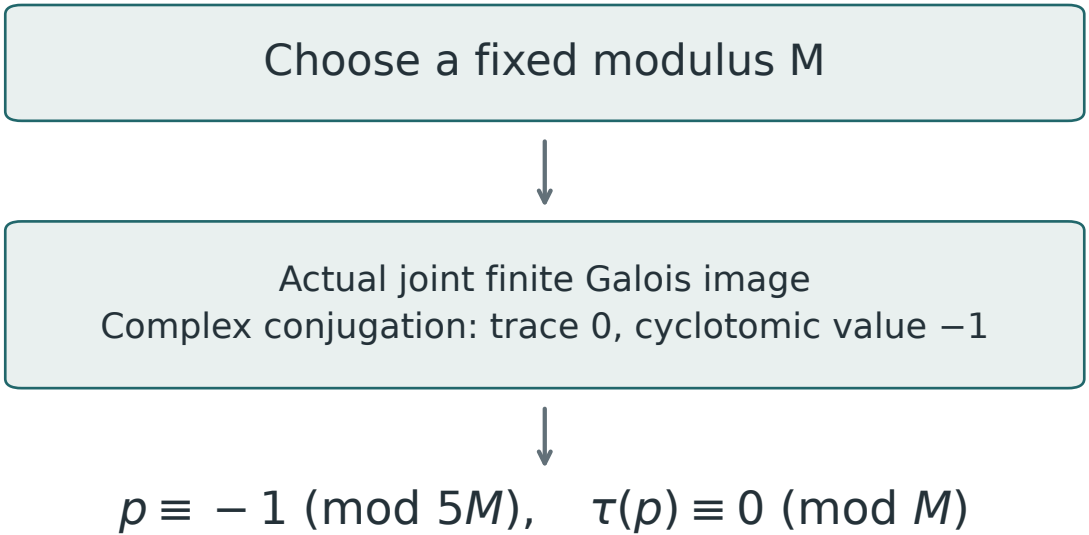
Choose a fixed modulus M
Actual joint finite Galois image
Complex conjugation: trace 0, cyclotomic value −1
$p \equiv -1 \pmod{5M}, \quad \tau(p) \equiv 0 \pmod{M}$